

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 13: Tropical Meteorology, Revolving Storms & Climatology

Syllabus Source: CAE Oxford (pp. 439-472) | Pass Target: $\geq 90\%$ AviationExam

1. THE INTERTROPICAL CONVERGENCE ZONE (ITCZ)

The ITCZ is the equatorial meteorological trough where the Northeast Trade Winds of the Northern Hemisphere meet the Southeast Trade Winds of the Southern Hemisphere:

- Characteristics: Intense solar heating causes broad-scale convergence and ascent, generating massive clusters of Cumulonimbus (CB) reaching FL 550 to FL 600. Known maritime name: The Doldrums.
- Seasonal Migration: Migrates north of the equator in Northern summer (July) and south in Northern winter (January). Migration is much greater over land masses (up to 25°N over Asia) than over oceans.

2. TROPICAL REVOLVING STORMS (TRS)

Severe tropical cyclones with warm cores and sustained winds ≥ 64 knots. Names: Hurricane (Atlantic/Eastern Pacific), Typhoon (Northwest Pacific), Cyclone (Indian Ocean/South Pacific):

- Mandatory Formation Conditions: 1. Warm ocean water with Sea Surface Temperature (SST) $\geq 26^{\circ}\text{C}$ to 27°C down to 50 m depth; 2. Latitude between 5° and 20° (Coriolis force is too weak to create rotation between 0° and 5° !); 3. Low vertical wind shear (< 10 kt); 4. Pre-existing tropical disturbance.
- TRS Anatomy: The Eye: 10 to 30 NM diameter central core with calm/light winds, descending air (subsidence), clear skies; The Eyewall: Ring of ferocious towering CBs surrounding the eye with MAXIMUM SUSTAINED WINDS (100 to 180 kt) and torrential rain; Storm Surge: Coastal sea level rise responsible for 90% of casualties.

3. THE ASIAN MONSOONS

- Southwest Monsoon (Summer: June to Sept): Intense heating over Asia creates a deep continental heat low. Warm, moisture-laden maritime air from the Indian Ocean sweeps northeast across India, bringing torrential rainfall and continuous low cloud.
- Northeast Monsoon (Winter: Dec to March): The massive Siberian High builds over cold Asia. Cool, dry continental air flows southwest towards the ocean, producing clear skies and dry weather.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Tropical Revolving Storms (TRS)

Formation Ingredients: - Sea Surface Temperature $\geq 27^{\circ}\text{C}$ with deep warm layer (50 m). - Distance from Equator ≥ 5 degrees latitude (Coriolis required to trigger spin; never on equator!). - Low vertical wind shear (< 15 kt) throughout troposphere. Dangerous vs Navigable Semi-Circle (Northern Hemisphere): - RIGHT-HAND Semi-Circle is the DANGEROUS semi-circle: Storm translational speed ADDS to rotational wind, and winds blow aircraft directly toward storm path. - LEFT-HAND Semi-Circle is NAVIGABLE: Forward speed subtracts from rotational wind.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why do Tropical Revolving Storms (Hurricanes / Typhoons) NEVER form between latitudes 0° and 5° North

- [A]: Ocean water is too cold
- [B]: The CORIOLIS FORCE is too weak near the equator to initiate cyclonic rotation
- [C]: Trade winds are too strong
- [D]: Too many islands

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Coriolis parameter is proportional to $\sin(\text{latitude})$. At latitudes under 5°, Coriolis force is virtually zero, preventing air from spinning into a closed vortex.

Q2: What weather conditions exist inside the 'EYE' of a mature Tropical Revolving Storm?

- [A]: Maximum hurricane-force winds
- [B]: CALM OR LIGHT WINDS, descending air (subsidence), and relatively CLEAR SKIES surrounded by the towering eyewall
- [C]: Continuous hail
- [D]: Severe turbulence

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Sinking air in the central eye suppresses cloud formation, creating an eerie pocket of calm air surrounded by violent eyewall winds.

Q3: What causes the wet SOUTHWEST MONSOON over the Indian subcontinent between June and September

- [A]: Cold fronts from Antarctica
- [B]: Intense summer solar heating forming a continental thermal low over Asia, drawing warm, moist maritime air from the Indian Ocean
- [C]: High pressure over the Himalayas
- [D]: Volcanic activity

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Asian thermal low draws equatorial maritime air across the Arabian Sea and Bay of Bengal, unleashing the summer monsoon.